

EEE4114F: Problem Set 4 (sampling)

1. The sequence

$$x[n] = \cos\left(\frac{\pi}{4}n\right), \quad -\infty < n < \infty$$

was obtained by sampling a continuous-time signal

$$x_c(t) = \cos(\Omega_0 t), \quad -\infty < t < \infty,$$

at a sampling rate of 1000 samples/s. What are two possible values of Ω_0 that could have resulted in the sequence $x[n]$?

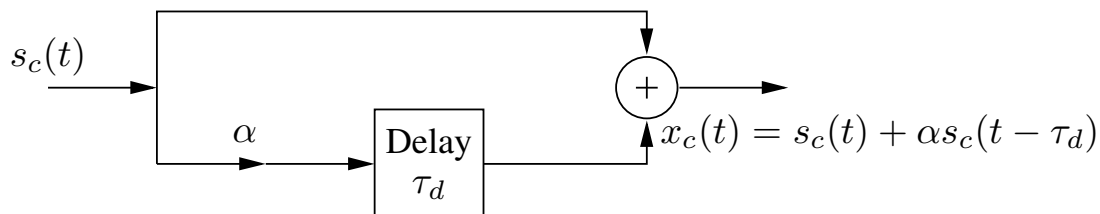
2. The continuous-time signal

$$x_c(t) = \sin(20\pi t) + \cos(40\pi t)$$

is sampled with a sampling period T to obtain the discrete-time signal

$$x[n] = \sin\left(\frac{\pi n}{5}\right) + \cos\left(\frac{2\pi n}{5}\right).$$

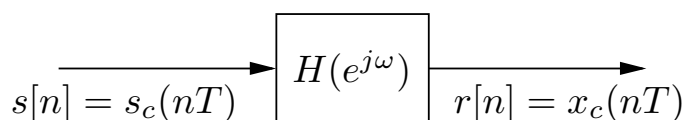
- (a) Determine a choice of T consistent with this information.
 - (b) Is your choice for T in part (a) unique? If so, explain why. If not, specify another choice of T consistent with the information given.
3. A simple model of a multipath communication channel is indicated below:



Assume that $s_c(t)$ is bandlimited such that $S_c(j\Omega) = 0$ for

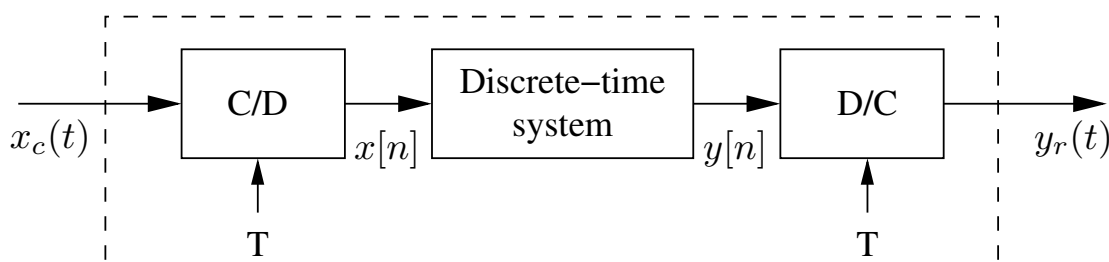
$|\Omega| \geq \pi/T$ and that $x_c(t)$ is sampled with a sampling period T to obtain the sequence $x[n] = x_c(nT)$.

- (a) Determine the Fourier transform of $x_c(t)$ and the Fourier transform of $x[n]$ in terms of $S_c(j\Omega)$.
- (b) We want to simulate the multipath system with a discrete-time system by choosing $H(e^{j\omega})$ in the figure so that the output $r[n] = x_c(nT)$ when the input is $s[n] = s_c(nT)$. Determine $H(e^{j\omega})$ in terms of T and τ_d .
- (c) Determine the impulse response $h[n]$ in



when (i) $\tau_d = T$ and (ii) $\tau_d = T/2$.

4. In the figure below,



assume that

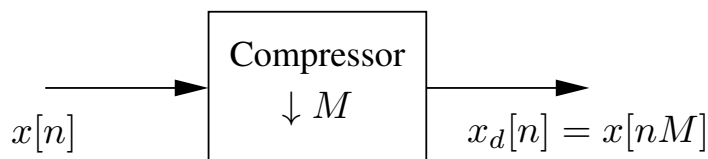
$$H(e^{j\omega}) = j\omega/T, \quad -\pi \leq \omega < \pi,$$

and $T = 1/10$ seconds.

- (a) For each of the following inputs $x_c(t)$, find the corresponding output $y_c(t)$.
 - i. $x_c(t) = \cos(6\pi t)$
 - ii. $x_c(t) = \cos(14\pi t)$
- (b) Are the outputs $y_c(t)$ those you would expect from a differentiator?

5. Which of the following signals can be downsampled by a factor of 2

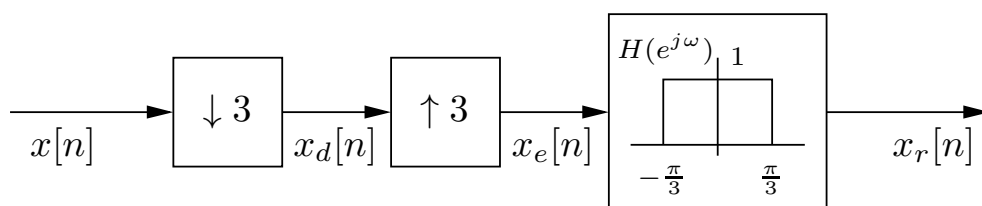
using the system



without any loss of information?

- (a) $x[n] = \delta[n - n_0]$, for n_0 some unknown integer
- (b) $x[n] = \cos(\pi n/4)$
- (c) $x[n] = \cos(\pi n/4) + \cos(3\pi n/4)$
- (d) $x[n] = \sin(\pi n/3)/(\pi n/3)$.
- (e) $x[n] = (-1)^n \sin(\pi n/3)/(\pi n/3)$.

6. Consider the system



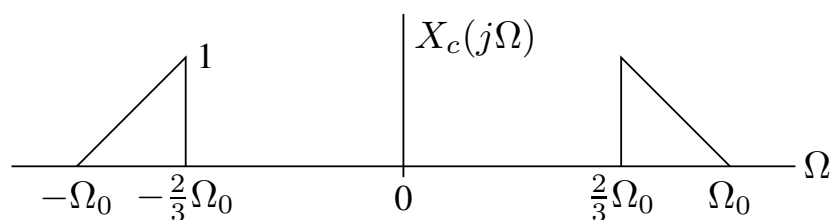
For each of the following input signals $x[n]$, indicate whether the output $x_r[n] = x[n]$.

- (a) $x[n] = \cos(\pi n/4)$
- (b) $x[n] = \cos(\pi n/2)$
- (c)

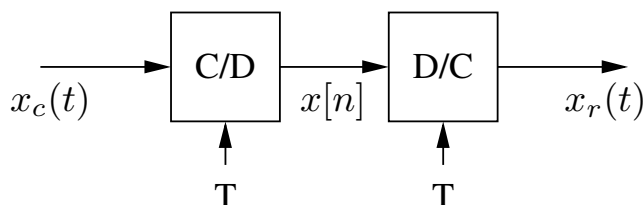
$$x[n] = \left[\frac{\sin(\pi n/8)}{\pi n} \right]^2$$

Hint: Use the modulation property of the Fourier transform to find $X(e^{j\omega})$.

7. The continuous-time signal $x_c(t)$ with the Fourier transform

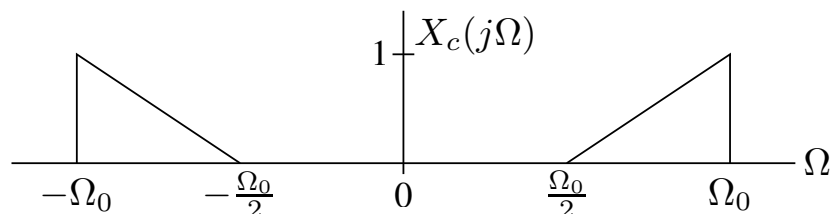


is passed through the system



Determine the range of values for T for which $x_r(t) = x_c(t)$.

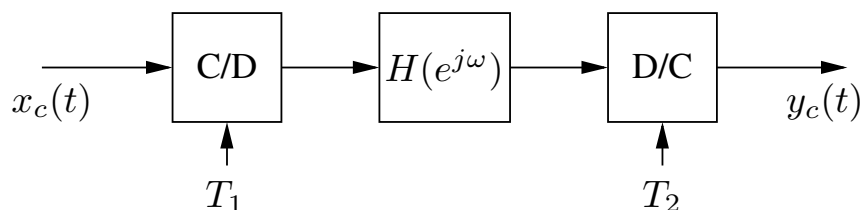
8. A continuous-time signal $x_c(t)$, with Fourier transform



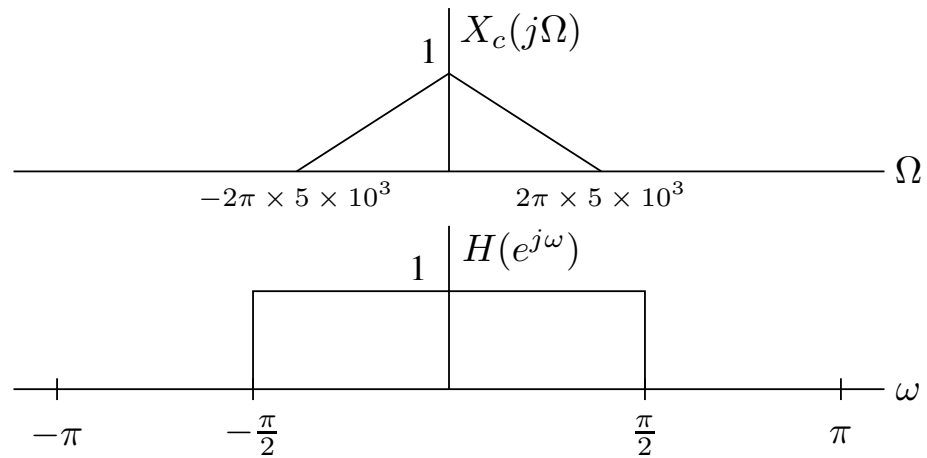
is sampled with sampling period $T = 2\pi/\Omega_0$ to form the sequence $x[n] = x_c(nT)$.

- Sketch the Fourier transform $X(e^{j\omega})$ for $|\omega| < \pi$.
- The signal $x[n]$ is to be transmitted across a digital channel. At the receiver, the original signal $x_c(t)$ must be recovered. Draw a block diagram of the recovery system and specify its characteristics. Assume that ideal filters are available.
- In terms of Ω_0 , for what range of values of T can $x_c(t)$ be recovered from $x[n]$?

9. In the system



$X_c(j\Omega)$ and $H(e^{j\omega})$ are as shown below:



Sketch and label the Fourier transform of $y_c(t)$ for each of the following cases:

- (a) $1/T_1 = 1/T_2 = 10^4$.
- (b) $1/T_1 = 1/T_2 = 2 \times 10^4$.
- (c) $1/T_1 = 2 \times 10^4, 1/T_2 = 10^4$.
- (d) $1/T_1 = 10^4, 1/T_2 = 2 \times 10^4$.